



# WELCOME!

Online Training Course  
Foundations in Mobile Phone Data (MPD) for Policy:  
Design and implementation for MPD Initiatives

[FLOWMINDER.ORG](https://flowminder.org)

[NOMMON](https://nommon.org)

[positium](https://positium.org)

# Course overview

- Five online training sessions:
  - Introduction to Planning an MPD Initiative
  - Policy Applications
  - Developing Partnerships and Negotiating Data Access
  - Data Pipeline Architecture
  - Data Quality and Characteristics
- Two self-paced topics:
  - Data governance and safeguards
  - Communicating in MPD initiatives



# Introduction to Planning an MPD Initiative

Session 1

FLOWMINDER.ORG

NOMMON

positium

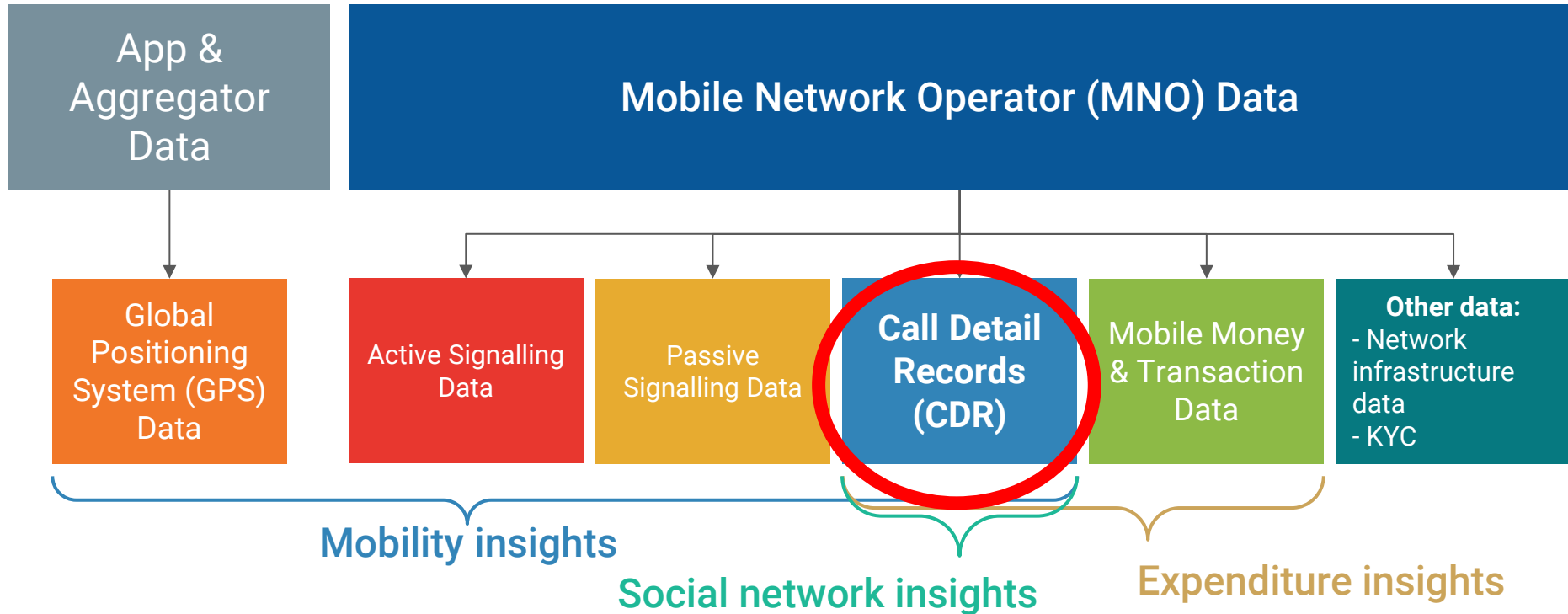
# Overview

This session covers the following topics:

1. Brief introduction to Mobile Phone Data (MPD) and Call Detail Records (CDRs)
2. **Reasons** for undertaking an MPD initiative
3. Key **requirements** for a successful MPD initiative
4. Key **stakeholders** to involve in an MPD initiative
5. Important **risks** to be considered when planning an MPD initiative
6. Designing **sustainable** MPD initiatives
7. **Tools** available for planning MPD initiatives

# What do we mean by 'mobile phone data'?

# Mobile Phone Data (MPD)



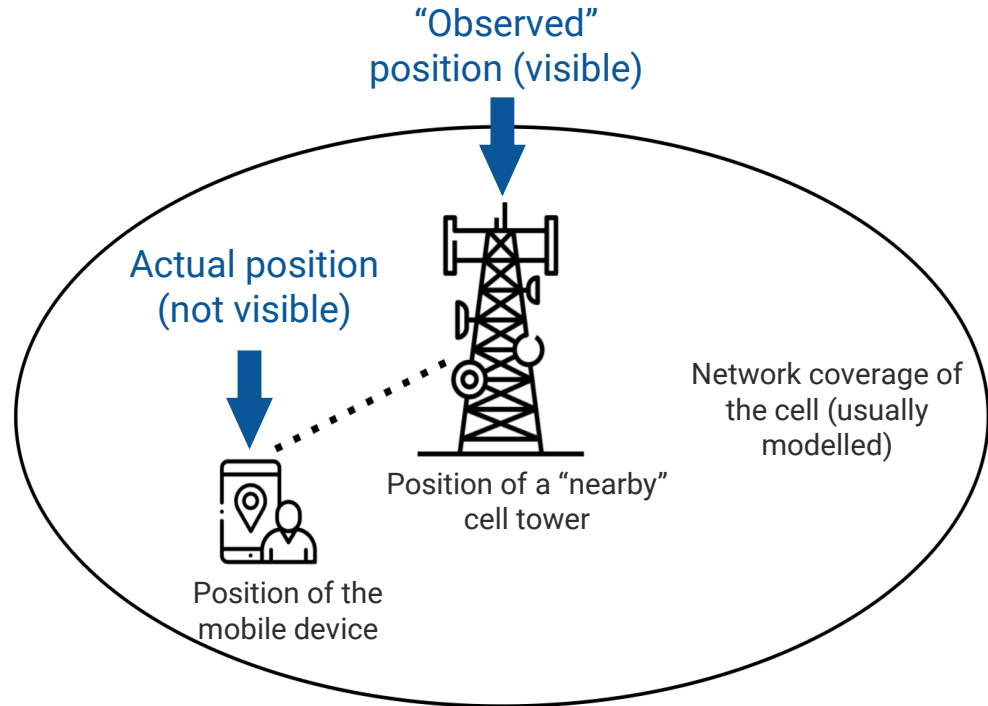
# GPS data vs Call Detail Record (CDR) data

|                 | Data collected /stored by | Spatial precision & resolution                            | Temporal resolution               | Population coverage         |
|-----------------|---------------------------|-----------------------------------------------------------|-----------------------------------|-----------------------------|
| <b>CDR data</b> | Mobile Network Operator   | Varies across areas<br>Approximate<br>Cell tower location | Intermittent<br>Depends on usage  | All phones                  |
| <b>GPS data</b> | Smartphone app            | Accurate<br>Exact<br>Device location                      | Continuous<br>Frequent & constant | App/OS users of smartphones |

# How CDR data is generated

## CDR data

- Passively generated when a subscriber
  - Makes or receives **a call**
  - Sends or receives **an SMS**
  - Uses mobile **data**
- Location is at **the cell tower** level
- Routinely stored by MNOs for billing purposes





# Key information contained in CDR data

| MSISDN          | CELL_ID   | TIMESTAMP           |
|-----------------|-----------|---------------------|
| AA204V1542DCA00 | 451154211 | 2016-10-10 15:35:25 |
| AA204V1542DCA01 | 451154211 | 2016-10-10 20:03:45 |
| AA204V1542DCA02 | 451154211 | 2016-10-10 21:56:15 |
| AA204V1542DCA03 | 451354312 | 2016-10-10 21:59:32 |
| AA204V1542DCA04 | 452616792 | 2016-10-10 22:42:23 |
| B45QHV45CAEVA5  | 476126941 | 2016-10-10 08:13:21 |
| B45QHV45CAEVA6  | 476126941 | 2016-10-10 08:14:15 |
| B45QHV45CAEVA7  | 476126941 | 2016-10-10 08:14:59 |
| B45QHV45CAEVA8  | 476126941 | 2016-10-10 12:41:01 |
| B45QHV45CAEVA9  | 476126941 | 2016-10-10 13:10:45 |
| B45QHV45CAEVA10 | 476126941 | 2016-10-10 15:20:43 |

Calling party identifier  
(pseudonymised)

Timestamp

Cell ID  
(location)

## Other variables in CDR data

- Receiving party identifier (pseudonymised)
- Network event type

CDR data do NOT contain information about the contents of a network event

# Reasons for planning an MPD initiative

# What motivates use of MPD, and specifically CDRs?

This data is an opportunity to **address important data gaps** and/or **augment traditional data sources**

Advantages over other data types:

- Produced in **real-time**
- **Passively generated**, so doesn't require active data collection (and associated costs)
- **Routinely recorded** and **stored** by MNOs, happens automatically as per of service usage by subscribers
- Large geographic **scale** (often national-level)

In low and middle income contexts, CDRs useful because:

- They are generated regardless of the type of mobile device
- Have higher penetration than smartphones

# Key requirements for MPD initiatives

# Key requirements for a successful MPD initiative

**Access to  
good quality  
data**

**Data storage  
and processing  
hardware**

**Software,  
tools and  
methods**

**Human  
resources  
with skills and  
availability**

**Clear, capable  
assigned  
leadership**

**Clear  
purpose  
e.g. policy  
application**

# Understanding the context

MPD initiatives need to consider a range of <sup>types of</sup> legislation and regulations:

- Data privacy and protection
- Communications and telecommunications
- Information and cybersecurity
- National security

May need to consider different jurisdictions, including **national**, **regional** and **supranational**, and broader legal landscape including penalties under **criminal law** for the misuse of personal data and how **constitutional rights** may provide grounds for legal challenges.

71%

Countries with data protection  
and privacy legislation  
(Apr 2022)

Secure expert legal advice on the  
use of MPD for statistical purposes

# Key stakeholders and their roles



Photo by [Jaime Lopes](#) on [Unsplash](#)

# Who are the key stakeholders involved in data access?

## Data Controller

- Mobile Network Operator(s)
- Telecoms Regulator

## Regulators

- Data Protection Authority
- Telecommunications Regulator
- National Security Agencies



## Data Users

- National Statistical Organisation
- Other Government Departments
- International Organisations
- Non-Governmental Organisations
- Private Sector

## Technical Service Provider(s)



# Role of statistical authority

---

NSO

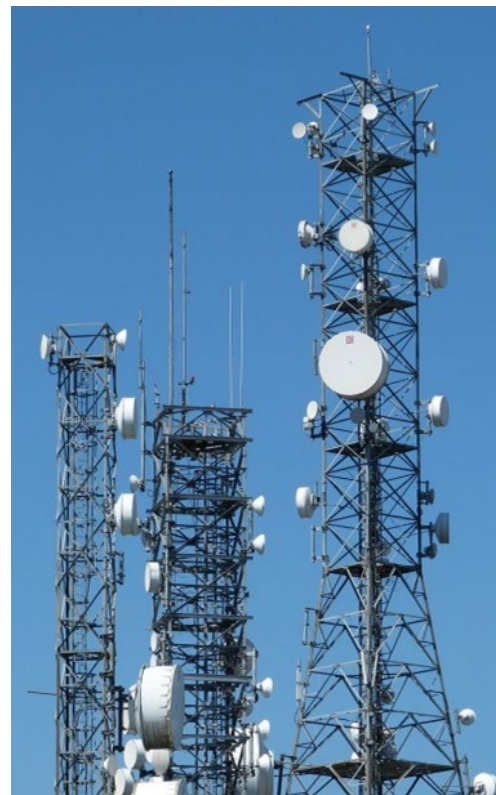
- Has legal **mandate** to produce official statistics (statistical law)
- Has legal basis to collect personal data (statistical **authority** )
- Has legal guarantees and **obligations** to protect it (statistical confidentiality)
- Has **experience** in statistical production
- Has access to **other data** sources for quality assurance and calibration

However:

- Capacity of the statistical authority varies between countries, so may need support and skills training

# Role of mobile network operator

- Data holder:
  - Original data source, can provide access to data
- Can leverage:
  - internal knowledge on the data source
  - technical capacity (IT)
- For ongoing, sustainable data pipelines, needs to provide:
  - Regular access to new CDR data, including all associated data fields
  - Regular updates to cell tower location data
  - Fixes when pipeline is disrupted (e.g. power outages, data gaps)
- Potential gains for MNO:
  - Benefit from statistical expertise of NSO
  - Gain access to other data sources at the NSO
  - Brand, reputation and corporate social responsibility
  - Financial incentives



# Role of telecommunications regulatory authority (TRA)

- Already has **regular contact** with MNOs
- Often has a **development agenda**
- Should be consulted to **ensure no objections** to data processing
- May already have mandate to collect records, so **can play different roles** :
  - Facilitate a partnership in the initial phases
  - Be the processor of Info Society statistics
  - Be the data gathering & processing party for other use cases also



Autorité de Régulation  
des Technologies de Communication



**INCM**

Autoridade Reguladora  
das Comunicações



Public Utilities  
Regulatory Authority  
*Equity in Development*



RWANDA  
UTILITIES  
REGULATORY  
AUTHORITY

# Role of data protection authority

- Sometimes obligatory by law to involve data protection regulatory body
- Can help strike balance between **public interest** and **privacy preservation**
- Can advise and guide **data protection measures** and whether they are proportional to the risks
- Can **enhance public trust** in an MPD initiative



# Other stakeholders

---

## ● Users

- Need to be engaged to specify their needs, can provide support (political, fundraising)

## ● Research community

- Provide academic rigour and resources for research

## ● Technical service providers

- Experts in the field, international exposure, best practices

## ● International organisations

- Provide funding, guide on standards, liaison between countries

## ● Legislators

- Sometimes a legislative approval or change is necessary

## ● Subscribers

- Are the data subjects and have an interest in protecting their privacy

# Different stakeholders have a range of concerns

| Subscribers                                                                                                        | Mobile Network Operator                                                                                                                                                                                                                                                                                     | Regulator                                                                                                                                                        | End-User                                                                                                              | Civil Society                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>■ Protect their privacy</li><li>■ Prevent the misuse of their data</li></ul> | <ul style="list-style-type: none"><li>■ Protect subscribers<ul style="list-style-type: none"><li>○ Legal responsibility</li><li>○ Reputation</li></ul></li><li>■ Protect commercially-sensitive data<ul style="list-style-type: none"><li>○ Cell tower locations</li><li>○ Market share</li></ul></li></ul> | <ul style="list-style-type: none"><li>■ Adherence to regulations<ul style="list-style-type: none"><li>○ e.g. protecting individuals' privacy</li></ul></li></ul> | <ul style="list-style-type: none"><li>■ Data is fit-for-purpose</li><li>■ Capacity to use the data products</li></ul> | <ul style="list-style-type: none"><li>■ That data is not misused<ul style="list-style-type: none"><li>○ Accident</li><li>○ Purposefully</li></ul></li><li>■ That individuals' rights are respected</li></ul> |

# Risks associated with MPD work

# Risks to be aware of in MPD initiatives

## ● Data-related risks:

- Privacy Risks
- Data Breaches and Security Risks

## ● Ethical Concerns

- Potential for bias, misleading results and ill-informed decision-making
- Potential for targeting of vulnerable groups (e.g. IDPs, migrants)

## ● Reputational and Trust Risks

- Public trust is important, public backlash can put a stop to initiatives
- Purpose and mandates are both important

## ● Regulatory and Legal Risks

- Compliance with laws and regulations



# Mitigation strategies

1. **Clear roles and responsibilities** for all staff related to data handling
2. **Training and capacity building** for staff
3. **Adopt risk management protocols** :
  - a. **Privacy enhancing techniques** (pseudonymisation, aggregation, redaction)
  - b. **Data encryption** (both stored and in transit)
  - c. **Strict authorisation and monitoring of access controls**
4. **Conduct security audits** to identify and address system vulnerabilities
5. **Assess compliance** with relevant data protection regulations; engage DPA for input
6. **Assess ethical considerations** ; involving others as appropriate (ethics boards, steering committees, academia/CSOs)

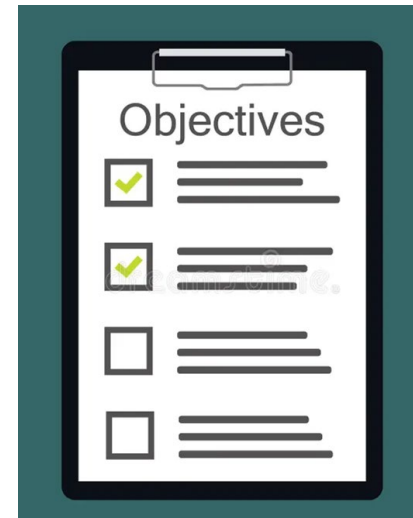
# Successful design for sustainability

# How to enable sustainable MPD initiatives

- Establish clear goals and objectives
- Ensure ethical use, legal compliance and data privacy
- Develop a robust data governance framework
- Design for sustainability in data collection and processing
- Invest in technical capacity and expertise
- Engage stakeholders and communicate effectively
- Integrate monitoring, evaluation and learning
- Identify investment requirements for resourcing

# Establish clear goals and objectives

- **Define the purpose** of accessing MPD from the outset. Clarify the **goal**, e.g:
  - Improving public policy
  - Improving crisis response
  - Enhancing statistical outputs
  - Strengthening service delivery
- Ensure that the data's **application** will be **meaningful**
- Align the **data request** with the purpose
- **Engage stakeholders:**
  - Align expectations and secure buy-in for the goals identified



# Ensure ethical use, legal compliance and data privacy

Respect and protect **data privacy** :

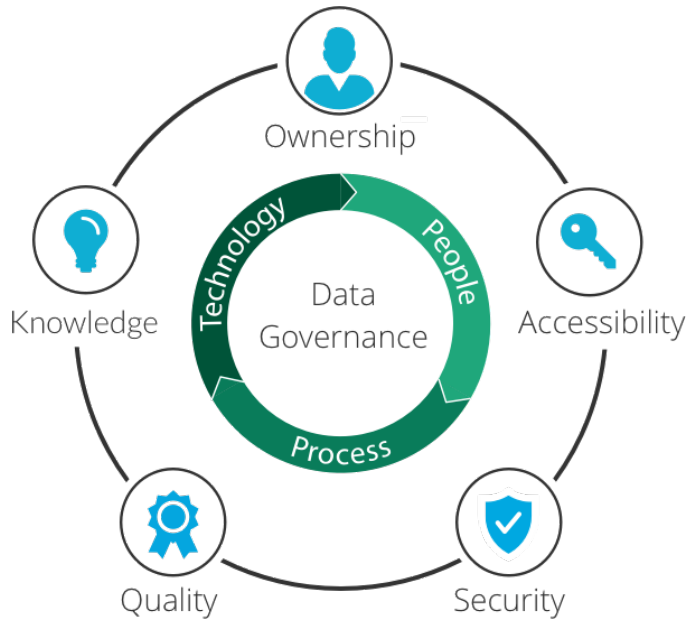
- Be clear about **local legal obligations** to ensure compliance
- Consider also local and international **norms and standards**

Adopt an **ethical** approach:

- Clarify early-on the **basis** for processing, any **ethical implications** and how to address these
- Be alert to actual or perceived **ethical risks** (unethical profiling, discrimination, surveillance)



# Develop a robust data governance framework



Create a data governance framework that defines:

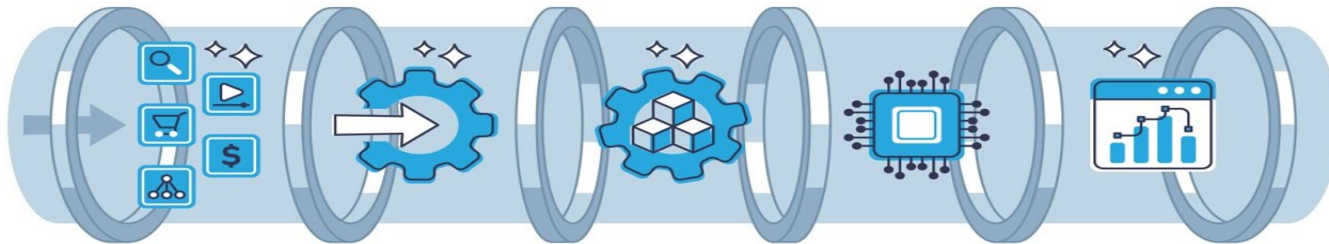
- Data stewardship roles
- Security measures
- Compliance protocols
- Accountability processes
- Data access control mechanisms
- Approval processes
- Quality assurance standards and approaches

# Design for sustainability in data collection and processing

**Scalable infrastructure:** As the volume of data grows, you'll need to be able to scale up data storage and processing power

**Continuous data flow:** Sustainable pipelines require long-term data access agreements

**Automate data pipelines** where feasible, to minimize human error and manual interventions



# Invest in technical capacity and expertise

## Deploy skilled personnel:

- Data engineers
- Data scientists
- Data analysts



## Delivery training and capacity -building:

- Conduct skills needs assessment and gap analysis to prioritise investments
- Hire in or build up key foundational skills for big data analytics
- Provide training in data privacy, security, and ethical data usage to all involved across partners



## Engage stakeholders and communicate effectively

- Engage all relevant key **stakeholders** early-on, identify any with concerns, and address them in the design
- Talk to the envisaged end users and conduct **user requirements gathering** early in the design process
- Develop a **clear dissemination plan for data products** , including for engaging civil society, and public-facing messaging
- Create channels for **feedback** to get any concerns or suggestions for improvement from stakeholders



# Integrate monitoring, evaluation and learning

## Monitor and evaluate impact

Put in place mechanisms to assess both progress and impact

Develop key performance indicators (KPIs) to measure success  
identify areas for improvement.

Be alert to changes in external environment like new technologies, changes to regulation, new/changed data sources and be adaptive

Refine and adapt the initiative based on findings and lessons learned.



# Identify investment requirements for resourcing

- Be clear about the relative **cost-benefit analysis** of deploying MPD for the chosen use case(s)
- **Set-up costs** are likely to be higher in the short-term but **maintenance costs** will accumulate over time.
- Cost structures will **depend on choices** about data access, data workflows and who does what in the data pipeline
- Identify not only what to fund but **how to fund** it:
  - In-kind contributions
  - Existing organisational budget allocations and/or project financing
  - Free resources e.g. open-source software
  - Donor funds



HUMAN RESOURCES



# Identify investment requirements for resourcing (cont)

## ● Budget items to consider:

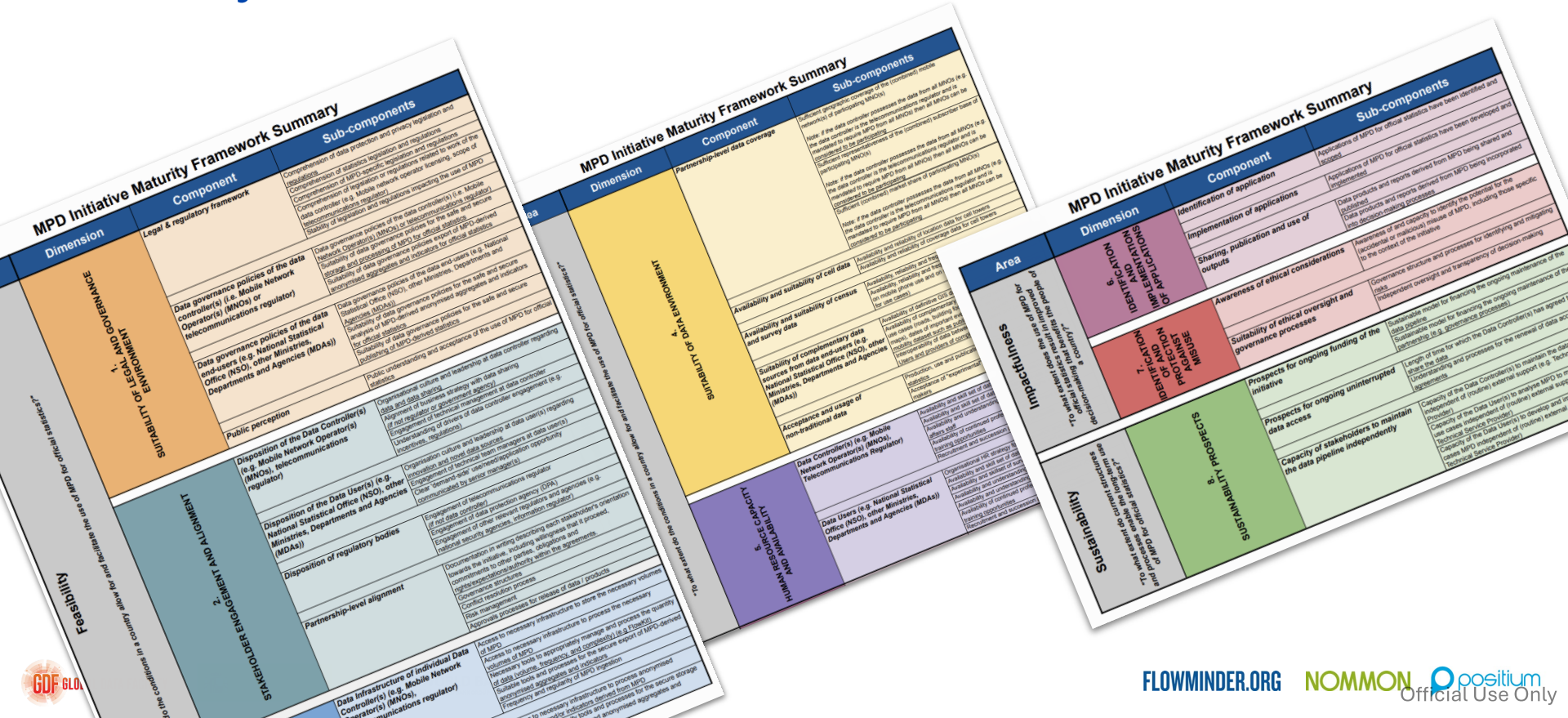
- **Data** – MNO data, survey data and other datasets
- **Skills** – Internal skill development and hiring
- **Technical assistance** – Sourcing external technical service providers for various stages of the project cycle
- **Infrastructure** – For data collection, ETL, processing, archiving and application delivery
- **Software** – Licences and open-source tools across the data pipeline
- **Workshops and travel** – National workshops for stakeholder engagement and dissemination, study visits, internships

# Additional Tools and Resources

# Theory of Change as a planning tool



# Maturity Assessment Framework & Tool



# Thank you!





The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).



Mobile Phone Data for Policy Programme (WB -GDF)  
© 2025 by Flowminder, Positium, Nommon, ITU and  
World Bank is licensed under CC BY 4.0.

To view a copy of this license, visit  
<https://creativecommons.org/licenses/by/4.0/>